

REQUIREMENT SPECIFICATION DOCUMENT

E-COMMERCE MANAGEMENT SYSTEM (EMS)

ONLINE GROCERY STORE

1. INTRODUCTION:

This section introduces the Online Grocery Store as a digital shopping platform that allows customers to purchase groceries, vegetables, fruits, dairy products, beverages, and household essentials online. The system simplifies grocery shopping by enabling users to browse products, place orders, make secure payments, and receive home delivery.

2. SCOPE:

The Online Grocery Store aims to achieve the following objectives:

- Customers can purchase grocery products online.
- Sellers can upload and manage grocery items.
- Orders and online payments are processed digitally.
- Customers can track their order status.
- The system supports multiple users and products as the business grows.

3. PROBLEM STATEMENT:

Traditional grocery shopping has several challenges:

- Customers spend time visiting supermarkets.
- Long waiting queues during billing.
- Limited product availability in nearby stores.

- Manual inventory management causes stock issues.
- Difficulty comparing prices and offers.

The Online Grocery Store solves these problems through digital shopping and automated order management.

4. STAKEHOLDERS:

The people who interact with the system include:

- Customers
 - Browse products, place orders, make payments, and track deliveries.
- Sellers
 - Upload products, update stock, and manage orders.
- Administrators
 - Manage customers, products, categories, and reports.
- Delivery Staff
 - Deliver orders and update delivery status.
- Payment Gateway
 - Process secure online transactions.

5. FUNCTIONAL REQUIREMENTS:

The system performs the following functions:

- User Registration and Login
 - Customers create accounts and log in securely.
- Product Browsing

- Customers browse groceries by category, brand, and price.
- Search Products
 - Search groceries using keywords.
- Shopping Cart
 - Add, update, or remove products before checkout.
- Order Placement
 - Place orders by selecting products and delivery address.
- Online Payment
 - Support UPI, debit card, credit card, net banking, and Cash on Delivery.
- Order Tracking
 - Customers can check Ordered, Packed, Shipped, Out for Delivery, and Delivered status.
- Product Management
 - Admin manages products, prices, categories, and stock.
- Customer Management
 - Manage customer accounts and purchase history.
- Report Generation
 - Generate sales, order, inventory, and customer reports.

6. NON-FUNCTIONAL REQUIREMENTS:

These define the quality of the system.

- Performance: Fast product loading and quick order processing.

- Security: Protect customer and payment information.
- Reliability: Continuous system availability with minimum downtime.
- Usability: Easy-to-use interface for all users.
- Scalability: Support increasing users and products.
- Availability: Available 24×7.
- Compatibility: Works on desktop, laptop, tablet, and mobile devices.
- Maintainability: Easy to update products and system features.

7. SYSTEM MODULES:

The system consists of the following modules:

- Customer Management
- Product Management
- Category Management
- Shopping Cart Management
- Order Management
- Payment Management
- Inventory Management
- Delivery Management
- Report Management

Each module performs a specific function within the system.

8. INPUT REQUIREMENTS:

The system requires the following inputs:

- Customer information
- Grocery product details
- Login credentials
- Delivery address
- Order details
- Payment information

These inputs are processed by the system.

9. OUTPUT REQUIREMENTS:

The system generates the following outputs:

- Customer account confirmation
- Product search results
- Order confirmation
- Payment receipt
- Order tracking details
- Sales reports
- Inventory reports

These outputs provide useful information to customers and administrators.

10. CONSTRAINTS:

The system has the following limitations:

- Requires internet connectivity.
- Payment gateway integration may depend on third-party services.
- Live delivery tracking may not always be available.
- Must comply with customer data privacy regulations.
- Product availability depends on stock levels.

CONCLUSION:

The Online Grocery Store provides a secure, reliable, and user-friendly e-commerce platform that makes grocery shopping faster and more convenient. It improves customer satisfaction while helping businesses efficiently manage products, orders, payments, and deliveries.

